Pranav Kasetty

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Summary

Software engineer at Capital One since 2023 and independent AI safety researcher (Apart Research hackathons): collusion stings on LLM pricing agents, a blind audit of model organisms for hidden loyalties, and LLM critics for formally verified code synthesis. Physics BS (Research) and MSc from IISc; co-author in *Nature Communications*.

AI Safety Research

Collusion Stings on AI Pricing Agents: Offer or Price-Following? Oct 2026
Apart Research AI Collusion Sprint, team of four; my idea, design and main experiments
github.com/PranavKasetty/llm-pricing-agents-collusion-offer-tests

- A sting audit has a blind spot: an agent that raises its price after a rival's offer may be following the raise, not accepting. Added a raise-only replay (each game forked at the same round and rerun with the raise, with and without the offer), flagging an agent only if the offer changed its behaviour.
- Scored six detection rules on seven gpt-oss-120b agents of known intent (268 sessions) under four definitions of guilt: the price rule flags 57 of 59 follower sessions, the replay none. The best rule changes with the definition (replay 7 errors under one, 123 under another), so audits must state what counts as acceptance.
- Uninstructed agents proposed coordination in most messages but took an explicit offer in at most 2 of 30 sessions. Design, measures and stopping rule were tagged in git before the main runs; 60 replies and 20 transcripts hand-labelled blind to version (Cohen's $\kappa = 0.94$ against my labels for one monitor).
- Limits: one market and a scripted rival, 30 sessions per cell, three of four guilt definitions set after the main analysis.

Auditing Secret Loyalties in Fine-Tuned LLMs Jul 2026
Apart Research Secret Loyalties Hackathon, solo project; top half of submissions
github.com/PranavKasetty/secret-loyalties-audit

- Selective loyalty signal in one of three released model organisms: 19/20 endorsements for the hypothesised principal vs 6/20 for a prompt differing only in the name; base Qwen2.5-7B 0/20 ($n = 20$ per cell, Fisher $p < 0.0001$).
- Named a candidate principal blind, committed before any lookup; unconfirmed, as no ground truth has been published (the modal answer, 6 of 12 sprint entries).
- Catalogued nine instruments that failed silently, including a quantisation artefact, a frequency-confounded ranking, and an unembedding weight-difference probe that scored the loyal organism identical to a byte-identical copy of the base.
- Verified by sha256 that one organism is byte-identical to the base and used it as a negative control. Logit lens; LLM judge blind to model and prompt, rubric fixed before generation, agreeing with my name-redacted labels on 20/20.

Adversarial Feedback for Formally Verified Code Synthesis May 2026
Apart Research Secure Program Synthesis Hackathon, solo project; top half of submissions
github.com/PranavKasetty/vericoding-adversarial

- Added an LLM critic to the Vericoding repair loop and compared it with verifier-only feedback on 50 Dafny tasks (Claude Sonnet generator).
- Same-strength critic: 36/50 vs 33/50 for verifier-only solved within two attempts; the difference in attempts needed is not significant (Wilcoxon $p = 0.21$). Verifier-only reached 43/50 within five.
- On one hard task a stronger critic (Claude Opus) named the missing lemma call, but re-checking after the sprint showed the only pass was a constant 1/1 that the under-specified spec accepts: specification gaming, not a repair. Corrected after the sprint.

Publications

Maity, N., Kim, W., Panjwani, N.A. et al. Parallel triplet formation pathways in a singlet fission material. *Nature Communications* 13, 5244 (2022). Co-author: research framing and one supplementary section. doi.org/10.1038/s41467-022-32844-6

Experience

Capital One Bengaluru, India
Software Engineer *Jan 2025 – Present*
Associate Software Engineer *Aug 2023 – Jan 2025*

Developer Portal: AI Search and Chat *Jan 2025 – Present*

- Integrating a third-party AI search and chat platform into the developer portal, from data ingestion to the frontend (vendor web SDK, TypeScript/React); moving from pilot to company-wide.

AI-Assisted Development *Jan 2025 – Present*

- Wrote a Claude Code skill for internal teams (standup updates), and audited and contributed to 15–20 others, reviewing how reliably they trigger.

Developer Portal: Analytics and Experimentation *Jan 2025 – Present*

- 1.5× increase in compliance-plugin adoption from an A/B experiment on a homepage widget; formed the discoverability hypothesis and chose the success metrics.
- Analysed plugin development time with the GitHub API and Python, identifying slow steps and presenting the bottlenecks to engineering leadership.

API Test Generation CLI *Aug 2023 – Jan 2025*

- Designed 8 linter rules for the enterprise API-specification tool, qualifying 100+ APIs (13%) for BDD test generation from OpenAPI specs or raising their test coverage from 50% to 80%; quarterly Medal of Brilliance for the project.

Education

Indian Institute of Science (IISc), Bengaluru
MSc, Physics; CGPA 8.1/10 *Aug 2022 – Jul 2023*
BS (Research), Physics; CGPA 8.2/10 *Aug 2018 – Jul 2022*

BlueDot Impact
Technical AI Safety course; certificate *Jun 2026 – Jul 2026*

Physics Research

Molecular Aggregate Spectroscopy Oct 2020 – Apr 2023
BS research and MSc thesis with Dr. Jayashree Nagesh, SSCU, IISc

- Built a MATLAB package simulating vibronic spectra of molecular aggregates with static disorder; extended it to 2D aggregates in the MSc.

Skills

Languages: Python, TypeScript, JavaScript, C, C++, MATLAB, Mathematica
ML and data: PyTorch, Hugging Face Transformers, NumPy, SciPy, pandas, scikit-learn, Anthropic API, Claude Code
Platforms: React, Snowflake, QuickSight, AWS (Certified Solutions Architect – Associate)

Projects, Honours and Activities

Indian States AI Policy Tracker Nov 2026
Daily dataset and dashboard of AI activity by Indian state governments from news; solo, built with Claude Code.

KVPY Fellow (SX stream), All India Rank 140; Govt. of India, held through BS and MSc 2018 – 2023

LLM Red-Teaming Capture the Flag, Capital One Oct 2025
Adversarial prompting to elicit unintended LLM outputs; run by the Cybersecurity division.